

Riemann Sums and Definite Integrals

ANSWER KEY



Approximating area with Riemann sums

- 1 Estimate $\int_0^4 x^2 dx$ using a right-endpoint Riemann sum with $n = 4$ subintervals.
 $\Delta x = 1$; right endpoints 1, 2, 3, 4: sum $= 1(1) + 1(4) + 1(9) + 1(16) = 30$.
- 2 Estimate $\int_1^5 (2x + 1) dx$ using a left-endpoint Riemann sum with $n = 4$ subintervals.
 $\Delta x = 1$; left endpoints 1, 2, 3, 4: values 3, 5, 7, 9: sum $= 24$.

The definite integral and its properties

- 3 Given $\int_0^3 f(x) dx = 8$ and $\int_3^7 f(x) dx = -2$, find:
a $\int_0^7 f(x) dx = 6$ **b** $\int_7^3 f(x) dx = 2$ **c** $\int_0^3 5f(x) dx = 40$
- 4 The graph of $f(x) = 4 - x^2$ is shown, together with the region between the curve and the x-axis on $[0, 2]$.
Evaluate $\int_0^2 (4 - x^2) dx$ exactly.
 $\left[4x - \frac{x^3}{3}\right]_0^2 = 8 - \frac{8}{3} = \frac{16}{3}$.
- 5 Evaluate $\int_{-2}^2 (x^3 - 4x) dx$ by recognizing the integrand's symmetry, without computing an antiderivative.
The integrand is odd ($f(-x) = -f(x)$), and the interval is symmetric about 0, so the integral is 0.

Midpoint and trapezoidal approximations

- 6 Estimate $\int_0^4 x^2 dx$ using a midpoint Riemann sum with $n = 4$ subintervals, and compare to the exact value $\frac{64}{3} \approx 21.33$.
 $\Delta x = 1$; midpoints 0.5, 1.5, 2.5, 3.5: sum $= 0.25 + 2.25 + 6.25 + 12.25 = 21$. Close to the exact value 21.33.
- 7 Estimate $\int_1^5 \frac{1}{x} dx$ using the trapezoidal rule with $n = 4$ subintervals, to three decimal places.
 $\Delta x = 1$, values at 1, 2, 3, 4, 5: 1, 0.5, 0.333, 0.25, 0.2. Trapezoidal:
 $\frac{1}{2}(1 + 2(0.5 + 0.333 + 0.25) + 0.2) = \frac{1}{2}(1.2 + 2.166) \approx 1.683$.

Signed area

- 8 The graph of $f(x) = x - 2$ crosses the x-axis at $x = 2$. Evaluate $\int_0^4 (x - 2) dx$ and explain why the result does not equal the total area between the curve and the axis.
 $\left[\frac{x^2}{2} - 2x\right]_0^4 = (8 - 8) - 0 = 0$. The integral gives signed (net) area: the region below the axis on $[0, 2]$ cancels the region above it on $[2, 4]$, even though both have positive area.

- 9 Find the total (unsigned) area between $f(x) = x - 2$ and the x -axis on $[0, 4]$, splitting the integral at the sign change $x = 2$.

$$\left| \int_0^2 (x - 2) dx \right| + \left| \int_2^4 (x - 2) dx \right|. \quad \int_0^2 (x - 2) dx = \left[\frac{x^2}{2} - 2x \right]_0^2 = (2 - 4) - 0 = -2, \text{ so } |-2| = 2.$$
$$\int_2^4 (x - 2) dx = \left[\frac{x^2}{2} - 2x \right]_2^4 = (8 - 8) - (2 - 4) = 0 - (-2) = 2. \text{ Total area} = 2 + 2 = 4.$$

Comparing Riemann sum estimates

- 10 For the increasing function $f(x) = x^2$ on $[0, 4]$, will the left-endpoint sum overestimate or underestimate the true area? Justify your answer.

Underestimate — since f is increasing, the left endpoint of each subinterval gives the smallest function value on that subinterval, so each rectangle sits below the curve.

- 11 Using the right-endpoint result (30) and left-endpoint result (24, computed earlier for a different function) is not directly comparable — instead, compare the right-endpoint estimate (30) for $\int_0^4 x^2 dx$ to the exact value $\frac{64}{3} \approx 21.33$, and explain the direction of the error.

The right-endpoint estimate (30) overestimates the true value (≈ 21.33), consistent with $f(x) = x^2$ being increasing — right endpoints give the largest function value on each subinterval, so the rectangles sit above the curve.